Name: ______________________

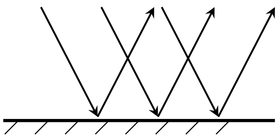
Date: ______________________

Mr. Rawson • Y8 MYP Physics

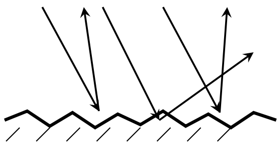
1.3 – The Law of Reflection

Unit 1: Light & EM Spectrum

Two kinds of reflection



smooth surface



rough surface

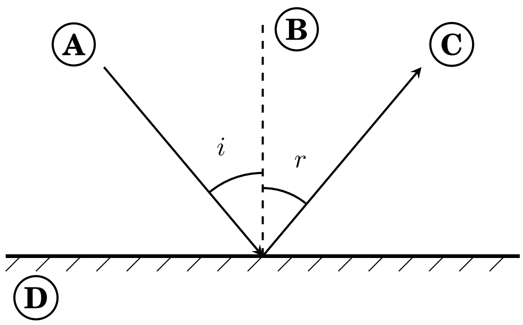
Light bounces off *every* surface it lands on. What changes from one surface to the next is not *whether* it reflects — it is **which way the reflected rays go**.

A smooth surface — specular reflection. A mirror is flat on the scale that matters to light. Rays that arrive parallel leave parallel, still in formation. The pattern survives the bounce, so your eye can rebuild it: you see an **image**.

A rough surface — diffuse reflection. Paper feels smooth but is not. Close up it is a landscape of bumps, each one facing a slightly different way. Rays that arrive parallel leave in all directions. The pattern is destroyed, so there is **no image** — but light still reaches your eye from every part of the page, which is exactly why you can see the paper itself.

The part people get wrong. Every one of those scattered rays still obeys the law of reflection *perfectly*. Diffuse reflection is not an exception to the rule. Each tiny bump has its own normal pointing its own way — so the same law gives a different answer at every point on the surface.

The four labels



A = incident ray   B = the normal   C = reflected ray   D = mirror

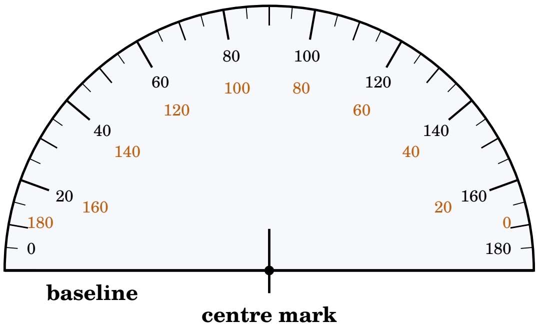
The **normal** is a construction line drawn at 90° to the mirror — always drawn **dashed**, because it is not a ray of light. *Both angles are measured from the normal, never from the mirror.*

One ray, and why we can draw it

The bulb sends light out in **every direction**. The **single slit** on the ray box blocks nearly all of it and lets one narrow strip through — so you get one clean straight line you can trace.

We draw that line with an arrow on it and call it a **ray**: a straight line showing the **direction** light travels. It is a model, not an object you could pick up.

Using a protractor



A protractor has **two scales**, counting opposite ways. Only one is right.

1. **Centre mark** on your dot.
2. **Baseline** along the normal — not the mirror.
3. Read the scale that starts at **0** on your line.

If it looks wrong, it probably is: wrong scale ⇒ you get $180°$ minus the true angle ($20°$ reads $160°$). Measured from the *mirror* ⇒ you get $90°$ minus the true angle ($20°$ reads $70°$).

The practical

Measure the law

Kit: ray box with a single slit, plane mirror, protractor, ruler, sharp pencil, plain paper.

1. Stand the mirror on the paper. Rule a line along the front of it — the **mirror line**.
2. Mark a **dot** on that line, near the middle.
3. With the protractor, rule a line at $90°$ to the mirror line through the dot. Make it **dashed** and label it **the normal**.
4. Shine the ray box so the ray hits **the dot**. Trace the ray going in and the ray coming out. **Both rays must pass through the dot.**
5. Measure **both** angles from the normal. Record them.
6. Move the ray box to a new angle and repeat.

Care: the lamp gets hot — carry the box by the case. Never shine it at anyone's eyes.

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Your five trials

Do each trial in its own box: mirror line, dot, dashed normal, then trace both rays through the dot. Write your two angles underneath.

Trial 1

angle of incidence $i =$ __________° angle of reflection $r =$ __________°

Trial 2

angle of incidence $i =$ __________° angle of reflection $r =$ __________°

Trial 3

angle of incidence $i =$ __________$^\circ$ angle of reflection $r =$ __________$^\circ$

Trial 4

angle of incidence $i =$ __________$^\circ$ angle of reflection $r =$ __________$^\circ$

Trial 5

angle of incidence $i =$ ________ $^\circ$ angle of reflection $r =$ ________ $^\circ$

Expected: in every trial r matches i to within a degree or two. A pair summing to 90° (e.g. 20 and 70) means the angle was measured from the mirror — remeasure from the normal. A reading over 90° means the wrong protractor scale was read.

Conclusions

1) State the law of reflection.

State: Give a specific name, value or other brief answer without explanation or calculation.

Model answer: The angle of incidence is equal to the angle of reflection, $i = r$, with both angles measured from the normal.

2) Describe the pattern in your results table.

Describe: Give a detailed account or picture of a situation, event, pattern or process.

Model answer: Whatever the angle of incidence, the angle of reflection is the same as it, to within a degree or two. As i increases from 20° to 60°, r increases with it, staying equal every time.

3) Explain why you can see your face in a mirror but not in a sheet of paper. Use the words *specular* and *diffuse*.

Explain: Give a detailed account including reasons or causes.

Model answer: A mirror is smooth, so reflection is specular: rays that arrive together leave together, and the pattern of light from your face is kept, making an image. Paper is rough, so reflection is diffuse: each ray hits a surface at a different angle and the rays scatter in all directions. The light still reflects — that is why you can see the paper — but the pattern is scrambled, so there is no image.

Extension

Predict, then test. If the angle of incidence goes up by $10°$, what happens to the angle *between* the incident ray and the reflected ray? And if you keep the ray still but tilt the *mirror* by $10°$ — what does the reflected ray do? Write your prediction, then try both with your kit.

The angle between the rays is $i + r = 2i$, so it grows by $20°$. Tilting the mirror by $10°$ swings the reflected ray by $20°$ — the normal tilts with the mirror, changing both i and r.

Criterion A — Knowing and understanding

This task is **formative**. It does not go on your report — we are *practising* the writing, and this is the standard we are practising against. Your three answers on the previous page (**State**, **Describe**, **Explain**) are the evidence.

Read down the right-hand column and notice what changes: the **verb** gets more demanding each time, and the situations go from **familiar** to **unfamiliar**.

Level	The student is able to:
0	Does not reach a standard indicated by any of the descriptors below.
1–2	i. **recall** scientific knowledge ii. apply scientific knowledge and understanding to **suggest solutions** to problems set in **familiar** situations iii. **apply** information to make judgments.
3–4	i. **state** scientific knowledge ii. apply scientific knowledge and understanding to **solve** problems set in **familiar** situations iii. **apply** information to make **scientifically supported** judgments.
5–6	i. **outline** scientific knowledge ii. apply scientific knowledge and understanding to solve problems set in familiar situations **and suggest solutions to problems set in unfamiliar situations** iii. **interpret** information to make scientifically supported judgments.
7–8	i. **describe** scientific knowledge ii. apply scientific knowledge and understanding to solve problems set in **familiar and unfamiliar** situations iii. **analyse** information to make scientifically supported judgments.

Wording taken verbatim from the *MYP Sciences guide* (2015), assessment criteria **Year 3**, Criterion A. Bold emphasis added to show what changes between levels.